SESSION ONE

To start saving for her home, Natalie has invested \$8000 in a special savings account that pays 6.0 % simple interest, with the interest being calculated annually.

(a) Calculate the amount of interest, correct to the nearest cent, earned each year.

Interest = 6 % of \$8000 = 0.06 × 8000 = \$480.00

(b) Using V_n to represent the value of the investment after n years, write a recurrence relation that models this investment situation.

$$V_0 = \$8000 \quad V_{n+1} = V_n + 480$$

(c) Use your calculator to determine recursively the value of Natalie's investment after four years. The table below has been given to assist you.

Year number	Calculation detail	Value of investment
0		8000.00
1	*8 000.00 + 480*	*8480.00*
2	*8 480.00 + 480*	*8960.00*
3	*8 960.00 + 480*	*9440.00*
4	*9 440.00 + 480*	*9920.00*

The value of the Natalie's investment after four years is \$9920.00.

Trent also started saving for his home. He started an account with \$2500 and each month added another \$150. The account pays interest at 6.0 % per annum, compounded monthly. The money is added to the account after the interest is calculated each month.

(d) Using V_n to represent the value of the investment after n years, write a recurrence relation that models this investment situation.

$$V_0 = \$2500, \quad V_{n+1} = 1.005V_n + 150$$

(e) Use your calculator to determine recursively the value of Trent's investment after six months. The table below has been given to assist you.

Year number	Calculation detail	Value of investment
0		2500.00
1	*2500 × 1.005 + 150*	*2662.50*
2	*2662.50 × 1.005 + 150*	*(2825.8125…) 2825.81*
3	*2825.8125… × 1.005 + 150*	*(2989.9415…) 2989.94*
4	*2989.9415… × 1.005 + 150*	*(3154.8912…) 3154.89*
5	*3154.8912… × 1.005 + 150*	*(3320.6657…) 3320.67*
6	*3320.6657… × 1.005 + 150*	*(3487.2690…) 3487.27*

The value of the investment after six months is $3487.27.

(e) What will be the value of Trent's investment after four years, correct to the nearest cent ?
Show details of any Finance Solver calculations in the table provided below.

N	**I(%)**	**PV**	**Pmt**	**FV**	**PpY, CpY**
48	*6.0*	*–2500*	*–150*	*0*	*12*

*Solve for **FV** : 11 290.90*

(f) Find the combined total value of Trent and Natalie's investments at the end of the sixth year of their investments.

Find Trent's total after six years

N	**I(%)**	**PV**	**Pmt**	**FV**	**PpY, CpY**
72	*6.0*	*–2500*	*–150*	*0*	*12*

*Solve for **FV** : 16 541.43905… ≈ $16 541.44*

Using $V_n = V_0 + nD$, *where* $V_0 = \$8000$, $n = 6$, $D = 480$

$$V_n = 8000 + 6 \times 480$$
$$= 10\,880$$

Natalie will have amassed $10 880.

Total saved = $16 541.44 + $10 880 = $27 421.44

SESSION TWO
Trent runs his own business from an office at the back of his house.

PROBLEM ONE
He recently purchased new furniture for his home office, spending a total of $8500.
He will depreciate the value of the furniture using prime cost or flat rate depreciation, where the annual depreciation will be 12.5% of the purchase price.

(a) By what amount, correct to the nearest cent, will the furniture be depreciated each year.

12.5% of $8500 = 0.125 × 8500 = $1062.50

(b) Using V_n to represent the value of the furniture after n years, write a recurrence relation that models this depreciation situation.

$V_0 = \$8500 \quad V_{n+1} = V_n - 1062.50$

(c) Use your calculator to determine recursively the value of the furniture, at the end of each year, for the first five years, and fill in the table below.

Year number	Calculation detail	Value of furniture
0		8500.00
1	*8500.00 – 1062.50*	*7437.50*
2	*7437.50 – 1062.50*	*6375.00*
3	*6375.00 – 1062.50*	*5312.50*
4	*5312.50 – 1062.50*	*4250.00*
5	*4250.00 – 1062.50*	*3187.50*

(d) Trent will replace this furniture when its value has depreciated below $3500. If he purchased the furniture at the end of 2013, at the end of which year will it reach its replacement value ?

If 2013 is year 0, and the value drops below $3500 in year 5, then the year in which it occurs is 2018.

PROBLEM TWO
He has a laptop computer which will need replacing soon. He bought it at the end of March 2014 for $4 400 and its value has been depreciating by 3% per month since then.

(a) Using V_n to represent the value of the computer after *n* months, write a recurrence relation that models this depreciation situation.

Recurrence relation : $V_0 = \$4400, \quad V_{n+1} = 0.97V_n$

Trent's accountant has advised him to trade in the computer for a new one when its value has dropped below a certain amount. Trent has determined that this value will be reached at the end of June 2016.

(b) What is the likely replacement value of the computer, written to two significant figures ?

Time to depreciate = March 2014 to June 2016 = 27 months

Using $V_n = R^n \times V_0$, *where* $R = 0.97$, $n = 27$, $V_0 = 4400$
$V_n = 0.97^{27} \times 4400 = \$1933.256\ldots \approx \$1933.26$

The depreciated value is approximately $1900 (to two sig. figs)

PROBLEM THREE
Trent purchased a new small car to use for his business at a cost of $38 000. He is allowed to depreciate it in value at a rate of 35 cents for every kilometre driven.

(f) Using V_n to represent the value of the car after travelling *n* kilometres, write a recurrence relation that models this depreciation situation.

$V_0 = \$38\,000 \qquad V_{n+1} = V_n - 0.35$

(g) What is the value of the car after it has travelled 22 500 kilometres ?

Using $V_n = V_0 + nD$, *where* $V_0 = \$1200$, $n = 22\,500$, $D = -0.35$
$V_n = 38\,000 - 0.35 \times 22\,500$
$= 30\,125$

(h) If the scrap value of the car is $15 000, how many kilometres will it have travelled, correct to the nearest hundred kilometre, to reach its scrap value ?

Using $V_n = V_0 + nD$, *where* $V_n = \$15\,000$, $V_0 = \$38\,000$, $D = -0.35$
$15\,000 = 38\,000 - 0.35n$
$0.35n = 23\,000$
$n = 65\,714.285\ldots$
The car will have travelled approximately 65 700 kilometres.

SESSION THREE

Trent and Natalie have found a house that they like, and to buy it they will have to borrow \$360 000 from their bank at 6.575 % per annum, repayable monthly over 20 years.

(a) What is the effective interest rate, correct to three significant figures, that Trent and Natalie will be paying on their loan at 6.575 % per annum compounded monthly ?

$$r_{effective} = \left(\left(1+\frac{6.575}{100\times 12}\right)^{12} - 1\right)\times 100\% = 6.77680\ldots \approx 6.78\ \%$$

(b) Show, using the Finance Solver function/app on your calculator that Trent and Natalie's monthly payment, correct to the nearest cent will be \$2699.98. Write the values used in the Finance Solver in the table below.

N	I(%)	PV	Pmt	FV	PpY, CpY
240	*6.575*	*360 000*	*0*	*0*	*12*

*Solve for **Pmt** : – \$2699.98*
The homebuyer pays \$2699.98 TO THE BANK each month

The bank has decided to make the monthly payment \$2700.00.

(c) Calculate the balance of Trent and Natalie's loan, correct to the nearest cent, after five years, if the monthly payments are \$2700.00. Write the values used in the Finance Solver in the table below.

N	I(%)	PV	Pmt	FV	PpY, CpY
60	*6.575*	*360 000*	*– 2700.00*	*0*	*12*

*Solve for **FV** : – \$308 485.11.*
The homebuyer still OWES THE BANK (negative sign) \$308 485.11.

(d) How much has been paid off the loan in five years ?

Principal reduced by (360 000 – 308 485.11) = \$51 514.89

(e) How much interest has been paid in five years ?

60 repayments were made of $2700.00 totalling $162 000.00.
Total interest paid is $162 000.00 – $51 514.89 = $110 485.11.

(f) What percentage, correct to the nearest whole number, does the interest represent of the total money paid after five years ?

$$\frac{110485.11}{162000} \times 100 = 68\ \%$$

(g) Approximately how much interest will be paid, correct to the nearest dollar, if the loan is amortised over the 20 years at the same interest rate (6.575 % p.a.) with the same monthly payment ($2700.00) ?

Interest = Total repaid – amount borrowed
= 20 × 12 × 2700 – 360 000
= $288 000

(h) In actual fact, the final repayment on Trent and Natalie's loan will be less than $2700, since their monthly payment as set by the bank is slightly higher than the actual required payment (see part (a)). Assuming that the interest rate does not change for the duration of their loan – remaining at 6.575 % p.a. – calculate correct to the nearest cent the amount of their final payment. Show details of any Finance Solver calculations in the table provided below.

N	**I(%)**	**PV**	**Pmt**	**FV**	**PpY, CpY**
240	*6.575*	*360 000*	*– 2700.00*	*0*	*12*

Solve for ***FV*** *: 8.62.*
The homebuyer over pays the bank $8.62.

Final payment = $2700 - $18.62 = $2691.38

Five years after taking out their loan, Trent and Natalie inherited \$80 000 from a relative and they decided to use it to help pay down their loan. After the payment was made, the bank offered two choices – continue to pay off the loan with the monthly repayment as previously calculated (and amortise the loan sooner), or continue to pay off the loan over the fifteen remaining years with a lower monthly repayment.

(i) Which option will cost Trent and Natalie the least money overall ?

OPTION ONE
Show details of any Finance Solver calculations in the table provided below.

work out FV with n = 60 first FV = \$308485.1103

N	I(%)	PV	Pmt	FV	PpY, CpY
???	*6.575*	*228 485.11*	*– 2700.00*	*0*	*12*

308485.1103 - 80000

Work out final payment by putting in N = 114 in calc and find FV

*Solve for **N** : 114*
A further 114 payments will be required to amortise the loan. with the final payment being a bit extra (\$41.85)

Total repaid = (60 + 114) × \$2700.00 + 41.85= \$469 841.85
Savings = 12 × 20 × \$2700 – \$469 841.85 = \$178158.15

OPTION TWO
Find the new repayment amount required.
Show details of any Finance Solver calculations in the table provided below.

N	I(%)	PV	Pmt	FV	PpY, CpY
180	*6.575*	*228 485.11*	*0*	*0*	*12*

*Solve for **Pmt** : – 1999.78*
The payment TO THE BANK (negative sign) is \$1999.78 per month.

Total repaid = 60 × \$2700.00 + 180 × \$1999.78 = \$521 960.40

Savings = 12 × 20 × \$2700 – \$521 960.40 = \$126 039.60

The greatest savings are made with the higher repayment amount paying the loan off sooner (less interest being paid !!)